

Enhancement of Wear Resistance of Hybrid Powder Coatings Using Nanoparticle Reinforcement

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Abstract

The experiments were designed using the Taguchi L9 orthogonal array to investigate the influence of key process parameters, namely percentage weight of TiO₂ nanoparticles, applied load, and sliding frequency, on wear rate and scratch hardness. Wear testing was conducted using a tribometer under dry sliding conditions, and scratch hardness was evaluated using a scratch tester.

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The results indicate that the addition of TiO₂ nanoparticles significantly improves wear resistance and hardness due to enhanced load-bearing capacity and formation of a protective tribolayer. An optimum nanoparticle content was observed, beyond which agglomeration adversely affected performance. Among the parameters studied, nanoparticle reinforcement showed the highest contribution to wear reduction, followed by applied load and frequency. The Taguchi method and ANOVA analysis effectively identified the optimal parameter combination for minimizing wear and improving coating performance.

The study concludes that TiO₂ reinforced hybrid powder coatings offer a promising solution for improving durability and wear resistance in engineering applications, particularly in automotive and industrial components.

1. Introduction

A coating is defined as a thin layer applied on the surface of a base material, commonly referred to as the substrate [1]. Coatings are primarily applied to enhance the appearance of the component or to provide protection against environmental and mechanical damage. Various techniques such as anodizing, powder coating, thermal spraying, and electro-deposition are used for coating deposition [2]. The curing process plays a critical role in developing proper adhesion between

the coating and the substrate. After curing, the coating attains the desired mechanical and chemical properties, which significantly improve the service life of the substrate [1-6].

Powder coating is a surface finishing process in which finely ground solid powder particles are deposited onto a substrate and subsequently cured by heating in an oven. During curing, the powder particles melt, flow, and chemically react to form a continuous, uniform, and durable coating layer. Powder coatings exhibit excellent resistance to abrasion, impact, scratching, and corrosion, along with good surface gloss. Due to lower material wastage, reduced labor requirements, energy efficiency, and long service life, powder coating is widely adopted across various industrial applications [7, 8].

In the early stages of development, coatings were applied mainly through flame spraying on metallic surfaces to protect them from wear and weathering. With technological advancement, the fluidized bed coating process was introduced, where preheated components were dipped into a powder bed. However, this technique often resulted in non-uniform coating thickness [9]. The introduction of electrostatic spray systems enabled coating of substrates at room temperature, ensuring better thickness control, uniform deposition, and efficient utilization of coating material.

Powder coating processes are extensively used in modern manufacturing industries to produce protective and functional surface finishes. These coatings provide enhanced durability, improved hardness, chemical resistance, and long-lasting surface gloss. Automation in powder coating systems allows their application on both hot and cold substrates, even under corrosive, high-pressure, and high-temperature operating conditions [10-14].

In the automotive sector, powder coatings are commonly used to protect under-hood components exposed to elevated temperatures and mechanical stresses. Additionally, powder coating improves the surface finish and durability of components such as wheels, mirror frames, oil filters, and coil springs. Automotive manufacturers increasingly employ powder coatings not only as primer layers but also as final topcoats due to their superior durability. Similarly, appliance manufacturers have replaced traditional energy-intensive porcelain coating processes with specially designed scratch-resistant powder coatings. As a result, components such as dryer drums, housings, and internal and external parts of microwave ovens are now widely powder coated [15-18].

2. MATERIALS AND METHODS

2.1 Work Material

Selection of the material for experimental work is one of the most important factors. There is much research in optimization of powder coating by optimizing process parameters but very few researches on the coating powder. Hybrid powder known for its good adhesion property and durability that's why they are mostly used in oil industries, refrigerators due to their good thermal conductivity it is also used in automobile industries. To increase the wear resistance of the hybrid coating nanoparticles are mixed which helps them to increase the mechanical property of the coating. Studies have been conducted for the homogenous mixing of the nanoparticles in the Hybrid composite coatings but very few researches are conducted on hybrid powder mixed with nanoparticles.

Titanium oxide nanoparticles

In this research TiO_2 nanoparticles are used as filler materials in the Hybrid powder coating. The properties of Hybrid powder was explained in the above chapter. Boron carbon Nanoparticles was provided by nano research lab tamil nadu having density 4.23 g/cm^3 , particular size 5-10 nm with melting point 1843°C . titanium- Hybrid Nano composites are mostly used in Nano composites. Research has been published about the effect of size and shape of nanofillers.



Figure 1: TiO_2 nanoparticles

2.2 Material Properties

Chemical composition

Table 1: Chemical composition

Element	Content
C- Carbon,	0.42-0.50 %
P- Phosphorus,	0.040 %
Si-Silicon	0.37 %
Mn- Manganese,	0.50-0.80 %
S- Sulphur,	0.50 %

Table 2: Mechanical property

Mechanical Properties	Metric	Imperial
Brinell Hardness,	170	170
Ultimate Tensile Strength,	565 MPa	81900 psi
Yield Tensile Strength,	310 MPa	45000 psi
Break at Elongation	16.0 %	16.0 %
Modulus of Elasticity	205 GPa	29700 ksi
Poissons Ratio	0.290	0.290
Shear Modulus	70.0 GPa	12600 ksi

2.3 Selections of Levels

Table 3 Levels of input parameters

Sr. No	Level 1	Level 2	Level 3
Reinforcement (%)	4	6	8
Load (N)	5	10	15
Frequency (Hz)	2	4	6

2.4 Standard L9 Orthogonal Array

A variety of standard orthogonal arrays are available in the Taguchi methodology, with each array designed to accommodate a specific number of factors and corresponding levels. The conventional notation used to represent an orthogonal array is $L_n(X^m)$, where n denotes the number of experimental runs, X represents the number of levels, and m indicates the number of factors.

Commonly used orthogonal arrays for reference include:

- Two-level arrays: $L_4 (2^3)$, $L_8 (2^7)$, $L_{12} (2^{11})$, $L_{16} (2^{15})$, $L_{32} (2^{31})$, $L_{64} (2^{63})$
- Three-level arrays: $L_9 (3^4)$, $L_{18} (2^1 \times 3^7)$, $L_{27} (3^{13})$, $L_{54} (2^1 \times 3^{25})$, $L_{81} (3^{40})$
- Four-level arrays: $L_{16} (4^5)$, $L_{32} (2^1 \times 4^9)$

Arrays such as $L_{18} (2^1 \times 3^7)$, $L_{54} (2^1 \times 3^{25})$, and $L_{32} (2^1 \times 4^9)$ are specifically designed for experiments involving mixed-level factors. In the present investigation, three factors-reinforcement content, applied load, and frequency-were considered, each at three levels. Accordingly, an L_9 orthogonal array (3^4) was selected to conduct the experimental trials.

Table 4 presents the selected L_9 orthogonal array, comprising a total of nine experimental runs, with each run corresponding to a unique combination of factor levels. In this study, the Taguchi L_9 design was implemented along with repeated trials to ensure experimental reliability, and multiple measurements were recorded for each experiment to improve data accuracy.

Table 4 Standards L9 orthogonal array

Sr. No.	Reinforcement (%)	Load (N)	Frequency (Hz)
1	4	5	2
2	4	10	4
3	4	15	6
4	6	5	4
5	6	10	6
6	6	15	2
7	8	5	6
8	8	10	2
9	8	15	4

3. Results and Discussion

The experimental investigation was conducted to evaluate the influence of key process parameters such as % weight of TiO₂ nanoparticles, applied load, and sliding frequency on the wear rate and scratch hardness of hybrid powder-coated mild steel substrates. The experiments were designed using the Taguchi L9 orthogonal array, and results were analyzed using Signal-to-Noise (S/N) ratio and ANOVA techniques

3.1 Experimental results for Wear Rate

Table 4 presents the L9 orthogonal array along with repeated response measurements for experimental runs one through nine. The repeated measurement approach was adopted to address the limitation of a saturated design in MINITAB 19 software. This table also indicates that the signal-to-noise (S/N) ratio values for repeated runs, such as run one and run ten, are identical, as the S/N ratio is computed based on the repeated response measurements. All S/N ratio calculations were performed using MINITAB 19.

Table 5 Cumulative data for S/N ratio Wear rate

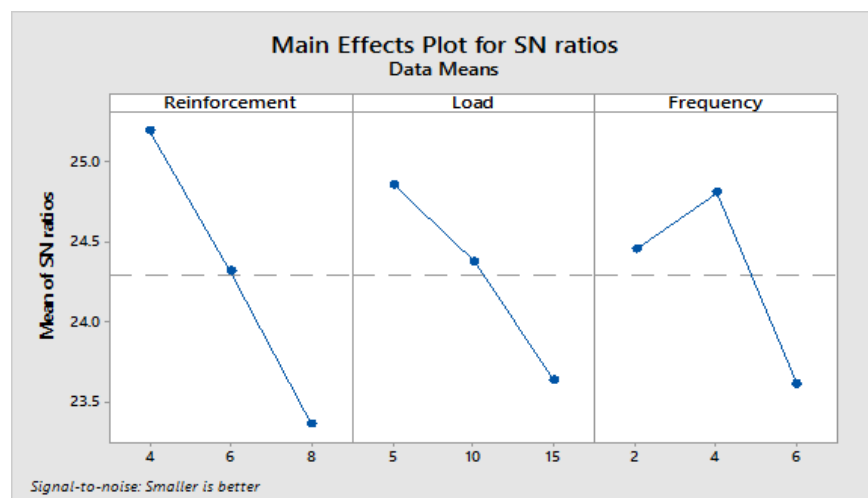
Trial No.	Reinforcement (%)	Load (N)	Frequency (Hz)	Wear Rate (mm ³ /Nm)	S/N Ratio
1	4	5	2	0.052	25.6799
2	4	10	4	0.047	26.5580
3	4	15	6	0.068	23.3498
4	6	5	4	0.057	24.8825
5	6	10	6	0.067	23.4785
6	6	15	2	0.059	24.5830
7	8	5	6	0.063	24.0132
8	8	10	2	0.070	23.0980
9	8	15	4	0.071	22.9748

3.2 Main effects of Wear Rate

The influence of each control factor reinforcement content, applied load, and frequency-on the wear rate was evaluated using the S/N ratio response table, which presents the S/N ratio corresponding to each level of the control factors. From the response graphs, the optimal parameter combination was identified as the condition at which the wear rate attains its minimum value and the coating exhibits improved performance. The analysis indicated that a reinforcement content of 2 wt.%, an applied load of 5 N, and a frequency of 4 Hz resulted in the lowest wear rate. A confirmation test was therefore conducted to verify the validity of the results obtained from the response graphs.

In this study, the higher-the-better S/N ratio criterion was adopted, and the S/N ratios for all experimental runs were calculated using MINITAB 19 software. It was observed that the optimum wear performance corresponded to the maximum S/N ratio values in the response graphs. The combination of control factor levels yielding the highest S/N ratio consistently provides optimal quality characteristics with minimal variability. The main effect plots clearly illustrate the variation in wear rate as the configuration of control factors changes from one level to another.

The main effect plots for the mean S/N ratio of wear rate are presented in Graph 1.



Graph 1 Main effect plots for mean of S/N ratio of Wear Rate

3.3 Analysis of variance (ANOVA)

In analysis of variance (ANOVA), the ratio of the variance associated with a process parameter to the variance attributed to experimental error is known as the Fisher ratio (F-value). This ratio is used to determine whether a particular parameter has a statistically significant effect on the selected quality characteristic. The significance of a parameter is assessed by comparing its calculated F-value with the corresponding critical F-value from standard F-distribution tables at a specified significance level (p-value). If the computed F-value exceeds the critical value, the parameter is

considered to have a significant influence on the response.

Table 6 ANOVA results of Wear Rate

Source	DF	Adj SS	Adj MS	F-Value	P-Value	% Contribution
Reinforcement	2	0.000230	0.000115	1.81	0.018	40.47
Load	2	0.000113	0.000056	0.89	0.008	20.03
Frequency	2	0.000198	0.000099	0.75	0007	35.10
Error	2	0.000024	0.000012			
Total	8	0.000564				

ANOVA serves as a statistical tool for validating experimental models and for testing the null hypothesis in planned experiments where multiple variables are evaluated simultaneously. By employing the F-test, ANOVA enables rapid assessment of variance contributions arising from different control factors. The results of this analysis are summarized in the ANOVA table, which presents the statistical significance and contribution of each factor.

The ANOVA results indicate that the p-values for all three control factors are less than 0.05, confirming that reinforcement content, applied load, and frequency significantly influence the wear behavior of the buffing wheel material. The cumulative contribution column in the ANOVA table represents the percentage contribution of each factor to the total variation, thereby indicating the relative impact of each parameter on the experimental outcome.

As shown in Table 6, reinforcement content (40.47%), applied load (20.03%), and frequency (35.10%) have a substantial effect on the wear rate. Among these, reinforcement exhibits the highest contribution, making it the most dominant parameter influencing wear rate, whereas load has the least influence among the three investigated factors.

3.4 Development of regression model for Wear Rate

A regression model was developed using MINITAB 19 software to predict the wear rate based on the selected process parameters. By substituting the experimentally obtained parameter values into the regression equation, the predicted wear rate was calculated for all levels of the study variables. The graphical comparison between the predicted and experimental wear rate values demonstrates a strong correlation, indicating the adequacy of the developed model.

To establish the relationship between the governing parameters-reinforcement content, applied load, and frequency-for the buffing wheel material, a multiple linear regression model was formulated using MINITAB 19. Only the parameters and terms found to be statistically significant were retained in the final regression equation. The resulting mathematical model was

then employed to estimate the wear rate, and regression analysis was conducted to validate the predictive capability of the model.

Regression Equation

Wear Rate = $0.0287 + 0.00308 \text{ Reinforcement} + 0.000867 \text{ Load} + 0.00142 \text{ Frequency}$ Table 7 gives comparison between experimentally measured and predicted Wear Rate by developed mathematical equation

Table 7 Experimental and predicted values of Wear Rate

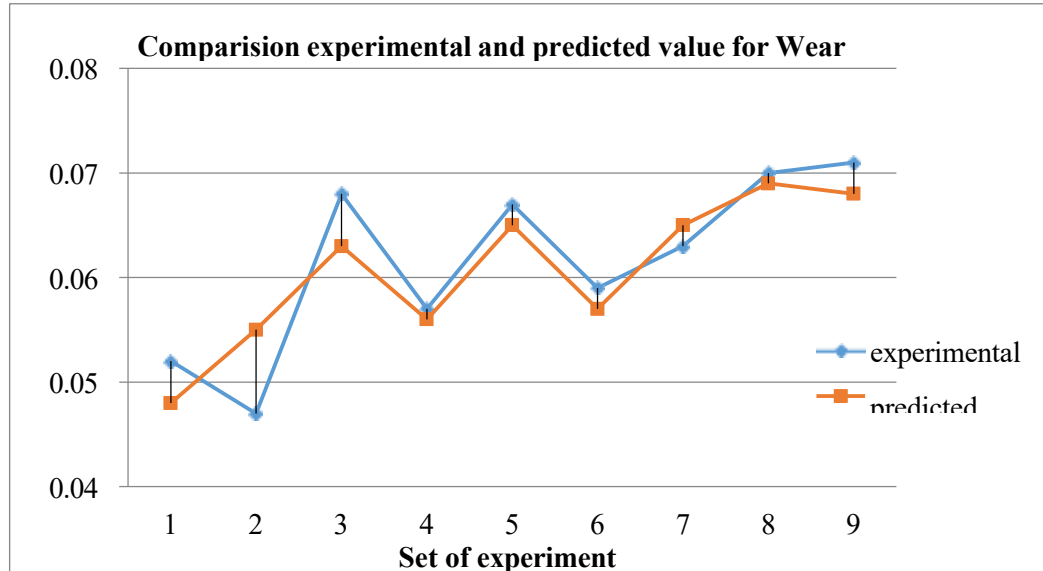
Sr. No.	Experimental value	Predicted value	Error %
1	0.052	0.048	4.56
2	0.047	0.055	2.59
3	0.068	0.063	3.58
4	0.057	0.056	2.58
5	0.067	0.065	3.05
6	0.059	0.057	7.28
7	0.063	0.065	6.27
8	0.070	0.069	3.89
9	0.071	0.068	4.78

The difference between the wear rate values predicted using the regression equation and the corresponding experimental results was found to be less than 10% for all experimental runs. This close agreement confirms the validity and reliability of the developed regression model. Graph 2 presents a graphical comparison between the experimentally measured wear rate values and those predicted by the regression equation.

The combined application of the Taguchi design method and ANOVA resulted in a mathematical model capable of predicting the response with reasonable accuracy. This level of agreement can be attributed to the fact that the present study considers the process parameters within a narrow and controlled range. Furthermore, significant efforts were made to maintain noise factors at constant levels throughout the experimental trials. The observed variation in wear rate decreasing for the initial experimental runs and increasing toward later runs-can be explained by the Taguchi L9 orthogonal array, in which each experiment from run 1 to run 9 corresponds to a unique combination of parameter levels. The comparison between the experimental and predicted wear rate values is illustrated in Graph 4.5.

Confirmation experiment result for Wear Rate

Table 8 shows the difference between value of Wear Rate of confirmation experiment and value predicted from regression model developed.



Graph 2 Comparison between experimental and predicted value of Wear Rate

Experiments were conducted for Reinforcement at level 1, Load at level 1 and Frequency at level 2

Table 10 Confirmation experiment result for Wear Rate

Parameter	Model value	Experimental value	Error %
Wear Rate (mm ³ /Nm)	0.051	0.046	9.80

A confirmation experiment was conducted by setting the process parameters at the optimal levels recommended by the Taguchi method. The wear rate obtained from this experiment was then compared with the value predicted by the regression model using the same parameter settings. The results show that the difference between the experimental and predicted wear rate values is 9.80%, indicating a good correlation between the measured and estimated results and thereby validating the effectiveness of the developed model.

4. CONCLUSIONS

4.1 Conclusion

In the present investigation, the effects of reinforcement content, applied load, and operating frequency on the wear behavior of hybrid powder coating were studied using Taguchi

optimization and ANOVA techniques to identify the optimum quantity of nanoparticles required for enhancing wear resistance. Based on the experimental results and statistical analysis, the following conclusions are drawn:

1. Experimental analysis revealed that the optimal parameter combination corresponds to 2 wt.% reinforcement, 5 N applied load, and 2 Hz frequency. Among all the reinforcement levels studied, 2 wt.% reinforcement exhibited superior wear resistance compared to both the unreinforced hybrid powder coating and the coating containing 10 wt.% reinforcement. ANOVA results further confirmed that reinforcement content contributed 40.47% to the overall variation in wear resistance, indicating its dominant influence.
2. ANOVA analysis demonstrated that frequency plays a significant role in controlling the wear rate. The percentage contributions of reinforcement, load, and frequency to the wear rate were found to be 40.47%, 20.02%, and 35.10%, respectively, confirming that all three parameters have a substantial impact on wear behavior.
3. A considerable improvement in wear resistance was observed for the coating containing 2 wt.% TiO₂ reinforcement compared to other compositions. The presence of nanoparticles reduces the effective contact area between the coating surface and the sliding pin, which delays the initiation of wear during the reciprocating abrasion test and leads to a lower wear rate.
4. It was observed that the addition of nanoparticles beyond 4 wt.% adversely affects the electrostatic charging efficiency of the powder during coating application. This reduction in charging efficiency negatively influences the mechanical properties of the deposited coating.
5. The hot-mixing technique employed for blending TiO₂ nanoparticles with the hybrid powder resulted in uniform dispersion of the nanoparticles. The incorporation of nanoparticles did not adversely affect the adhesion strength of the coating.
6. The confirmation experiment, conducted using the optimal parameter settings suggested by the Taguchi method, resulted in a lower wear rate, thereby validating the effectiveness of the selected process conditions for producing a high-quality hybrid coating reinforced with TiO₂ nanoparticles.
7. The wear rate values predicted by the regression model showed close agreement with the experimental results, with an error margin of less than 10%. This confirms the validity of the developed regression model, which can be effectively used to estimate wear rate within the

selected range of operating parameters.

4.2 Future scope

Hybrid powder coatings are well suited for applications involving highly corrosive environments and conditions where thermal stability is critical for prolonging coating service life. While the present study focuses on improving the wear resistance of hybrid powder coatings through nanoparticle reinforcement, several avenues remain open for further investigation. The following scope for future work is proposed:

1. The incorporation of alternative nanoparticles can be explored to evaluate their influence on the mechanical behavior and overall performance of the coating, along with detailed material characterization.
2. Microstructural analysis of the coated surfaces can be conducted to examine the dispersion, distribution, and interfacial bonding of nanoparticles within the coating matrix.
3. Development of hydrophobic or superhydrophobic surface layers on the coating can be investigated to enhance corrosion resistance and self-cleaning properties.
4. A comparative study of various homogeneous dispersion techniques, such as solution-based mixing, melt blending, and hot-mixing methods, can be performed to identify the most effective approach for achieving improved mechanical properties in powder coatings.
5. The effect of nanoparticle addition on surface roughness and its relationship with tribological performance can be examined in detail.
6. The study may be extended by varying the weight percentage of reinforcement beyond the current range and by using combinations of different nanoparticles to achieve synergistic improvements in coating performance.
7. Further research can involve repeating the experiments under different combinations of operating parameters, as well as conducting impact and fatigue tests to evaluate the impact strength and fatigue life of the reinforced composite coatings.

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